

U.G.C. NET EX COMPUTER (Solved with Expl

1. The definitions in an XML document are said to be _____ when the tagging system and definitions in the DTD are all in compliance.
- (a) well-formed (b) reasonable
(c) valid (d) logical

Ans. (c) : A DTD is document type definition. A DTD defines the structure and the legal elements and attributes of an XML document. Why use a DTD— with a DTD, independent groups of people can agree on standard DTD for interchanging data. An application can use a DTD to verify that XML data is valid.

2. Consider the JavaScript Code :

```
var y= "12";  
function f( ) {  
    var y= "6";  
    alert (this.y);  
    function g( ) {alert (y); }  
    g( );  
}
```

If M is the number of alert dialog boxes generated by this JavaScript code and D1, D2, ..., DM represents the content displayed in each of the M dialog boxes, then :

- (a) M=3; D1 displays "12"; D2 displays "6"; D3 displays "12".
(b) M=3; D1 displays "6"; D2 displays "12"; D3 displays "6".
(c) M=2; D1 displays "6"; D2 displays "12".
(d) M=2; D1 displays "12"; D2 displays "6".

Ans. (d) : Alert box will be called 2 times so M = 2. First dialog box alert (this.y) is used to access the value of variable which is outside from

f () . So D1 = 12

Second dialog box alert (y) inside nested function g () is used to access the local variable y. So D2 will be 6.

3. What is the output of the following JAVA program ?

```
class simple  
{  
    public static void main(String[ ] args)  
    {  
        simple obj = new simple( );  
        obj.start( );  
    }  
    void start( )
```

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```

{
    long [ ] P= {3, 4, 5};
    long [ ] Q= method (P);
    System.out.print (P[0]+P[1]+P[2]+":");
    System.out.print (Q[0] + Q[1] + Q[2]);
}
long [ ] method (long [ ] R)
{
    R [1]=7;
    return R;
}
} //end of class
(a) 12 : 15                (b) 15 : 12
(c) 12 : 12                (d) 15 : 15

```

Ans. (d) : Output is 15:15

1. p = 3, 4, 5

2. q = 3, 7, 5

When we have the statement `long [ ] Q = method (p).` It will call method function and p is passed as parameter and in method function we are changing `r [1] = 7.`

Actually arrays are passed as parameter and in method function we are changing `r[1] = 7.`

Actually arrays are passed as reference. So whatever changes you make in r, it actually gets reflected in p. Hence both the arrays will display sum = 3+7+5 = 15.

4. What is the output of the following ‘C’ program ? (Assuming little - endian representation of multi-byte data in which Least Significant Byte (LSB) is stored at the lowest memory address.)

```

#include <stdio.h>
#include <stdlib.h>
/* Assume short int occupies two bytes of storage
*/
int main ( )
{
    union saving
    {
        short int one;
        char two[2];
    };
    union saving m;
    m.two [0] = 5;
    m.two [1] = 2;
    printf(“%d, %d, %d\n”, m.two [0], m.two
    [1], m.one);
}/* end of main */

```

(a) 5, 2, 1282

(b) 5, 2, 52

(c) 5, 2, 25

(d) 5, 2, 517

Ans. (d) : Big-endian and little-endian are terms that describe the order in which a sequence of bytes are stored in computer memory.

Big-endian is an order in which the "big and" (most significant value in the sequence) is stored first (at the lowest storage address). Little-endian is an order in which the "little end" (least · significant value in the sequence) is stored first.

Short in tone; }
Char two [2]; } – 2 bytes

in union saving m; m is variable

here m·two [0] = 5; the 5 is not in single inverted comma ' ' so it is numeric value or number

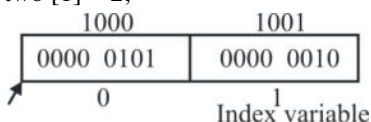
So, 5 binary equivalent = 0000 0101

and 2 binary equivalent = 0000 0010

Now we are using %d for printing the character.

m·two [0] = 5;

m·two [1] = 2;



incase of array "1" address will 1000 and 2nd will 1001.

% d will print decimal value of ·m· two [0] = 5; = 5

Now m· two [1] = 2; = 2

Now m· one



So, according to ·little· endian definition

0000 0101 will – Lowest memory address

0000 0010 – most significant byte.

Now evaluate this have value–

0000·0010 0000·0101
 29 28 27 22 21 20

Total will be = 517

5. Given below are three implementations of the swap() function in C++ :

(a)	(b)	(c)
<pre>void swap (int a, int b) { int temp; temp = a; a = b; b = temp; } int main() { int p=0, q=1; swap (p, q); }</pre>	<pre>void swap (int &a, int &b) { int temp; temp = a; a = b; b = temp; } int main() { int p = 0, q = 1; swap (p, q); }</pre>	<pre>void swap (int *a, int *b) { int *temp; temp = a; a = b; b = temp; } int main() { int p = 0, q = 1; swap (&p, &q); }</pre>

Which of these would actually swap the contents of the two integer variables p and q ?

- (a) (a) only (b) (b) only
(c) (c) only (d) (b) and (c) only

Ans. (b) : Void swap 2(int &a, int &b)

You are expecting the variable to pass by reference (which is the address of an integer variable).

So,

```
void swap (int &a, int &b)
{
    int temp;
    temp = a;
    a = b;
    b = temp;
}

int main( )
{
    int p = 0, q = 1; swap (p, q);
}
```

6. In Java, which of the following statements is/are True ?

S1 : The 'final' keyword applied to a class definition prevents the class from being extended through derivation.

S2 : A class can only inherit one class but can implement multiple interfaces.

S3 : Java permits a class to replace the implementation of a method that it has inherited. It is called method overloading.

Code :

- (a) S1 and S2 only
(b) S1 and S3 only
(c) S2 and S3 only
(d) All of S1, S2 and S3

Ans. (a) : In Java, items with the final modifier cannot be changed!

This includes final classes, final variables, and final methods.

A final class cannot be extended by any other class

A final variable cannot be reassigned another value

A final method cannot be overridden

A final class is simply a class that can't be extended.

(This does not mean that all reference to objects of the class would act if they were declared as final).

⇒ A Java class only extend one parent class. Multiple inheritance (extends) is not allowed. Interfaces are not classes, however, and a class can implement more than one interface.

The parent interfaces are declared in a comma-separated list, after the implements key words. In conclusion, yes, it is possible to do.

7. Which of the following statements is/are True ?

P : C programming language has a weak type system with static types.

Q : Java programming language has a strong type system with static types.

Code :

- (a) P only (b) Q only
(c) Both P and Q (d) Neither P nor Q

Ans. (c) : C programming language has a weak type system with static types. True

Java programming language has a strong type system with static types. True

Both Statements are True.

- 8. A graphic display system has a frame buffer that is 640 pixels wide, 480 pixels high and 1 bit of color depth. If the access time for each pixel on the average is 200 nanoseconds, then the refresh rate of this frame buffer is approximately :**

- (a) 16 frames per second
(b) 19 frames per second
(c) 21 frames per second
(d) 23 frames per second

Ans. (a) : Here 640 pixel wide of screen means vertical line on screen and 480 pixels high is means vertical line on screen.

Now width (w) = 640 px, weight (w) = 480 px

Total = $640 \times 480 = 307200$

Color depth = 1-bit/px

Memory required for total pixel = $w \times h \times \text{color depth}$
 $= 640 \times 480 \times 1\text{-bit} = 307200$

1 pixel takes 200 nano second to refresh

To refresh the whole screen = $640 \times 480 \times 200\text{ns}$
 $= 61440000 \text{ nano second}$

For 1 nano second = $\frac{1}{61440000}$ screen

Now convert nano second to second = 10^9

For 1 nano second = $\frac{1}{61440000} \times 10^9$
 $= 16.2760 = 16 \text{ (Approximately)}$

- 9. Which of the following statements is/are True regarding the solution to the visibility problem in 3D graphics ?**

S1 : The Painter's algorithm sorts polygons by depth and then paints (scan - converts) each Polygon on to the screen starting with the most nearest polygon.

S2 : Backface Culling refers to eliminating geometry with backfacing normals.

Code :

- (a) S1 only (b) S2 only
(c) Both S1 and S2 (d) Neither S1 nor S2

Ans. (b) : Backface Culling refers to eliminating geometry with backfacing normals. True

10. Consider the matrix $M = \begin{bmatrix} 2 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ representing

a set of planar (2D) geometric transformations in homogeneous coordinates. Which of the following statements about the matrix M is True ?

- (a) M represents first, a scaling of vector (2, 1) followed by translation of vector (1, 1)
- (b) M represents first, a translation of vector (1, 1) followed by scaling of vector (2, 1)
- (c) M represents first, a scaling of vector (3, 1) followed by shearing of parameters (-1, 1)
- (d) M represents first, a shearing of parameters (-1, 1) followed by scaling of vector (3, 1)

Ans. (a) : M represents first, a scaling of vector (2, 1) followed by translation of vector (1, 1)

Matrix for scaling vector (2, 1) =

2	0	0
0	1	0
0	0	1

Matrix for translation vector (1, 1) =

1	0	1
0	1	1
0	0	1

Multiplying both we get the resultant vector

2	0	2
0	1	1
0	0	1

M represents first, a scaling of vector (2, 1) followed by translation of vector (1, 1)

So, option (a) is correct.

11. Assume the following regarding the development of a software system P :

- Estimated lines of code of P : 33, 480 LOC
- Average productivity for P : 620 LOC per person-month
- Number of software developers : 6
- Average salary of a software developer : ₹ 50,000 per month

If E, D and C are the estimated development effort (in person-months), estimated development time (in months), and estimated development cost (in ₹ Lac) respectively, then (E, D, C) =_____.

- (a) (48, 8, 24) (b) (54, 9, 27)
- (c) (60, 10, 30) (d) (42, 7, 21)

Ans. (b) : Each person developed 620 LOC in a month
 So, 6 person will develop LOC = $620 \times 6 = 3720$ LOC in a month

$$\text{So, time} = \frac{.33480}{3720} = 9 \text{ month}$$

Cast (in lacks) = $.6 * .9 * 50,000 = 27$ Lac

So, two parameter will match so, option is (b).

12. Match the following in Software Engineering :

List - I		List - II	
a.	Product Complexity	i.	Software Requirements Definition
b.	Structured System Analysis	ii.	Software Design
c.	Coupling and Cohesion	iii.	Validation Technique
d.	Symbolic Execution	iv.	Software Cost Estimation

Code :

	(a)	(b)	(c)	(d)
(a)	(ii)	(iii)	(iv)	(i)
(b)	(iii)	(i)	(iv)	(ii)
(c)	(iv)	(i)	(ii)	(iii)
(d)	(iii)	(iv)	(i)	(ii)

Ans. (c) :

13. Which one of the following is not typically provided by Source Code Management Software ?

- (a) Synchronisation
- (b) Versioning and Revision history
- (c) Syntax highlighting
- (d) Project forking

Ans. (c) : Source code management software provides—

Synchronization, project forking, versioning, revision history.

But Syntax highlighting is not part of source code management software.

14. A software system crashed 20 times in the year 2017 and for each crash, it took 2 minutes to restart. Approximately, what was the software availability in that year ?

- (a) 96.9924%
- (b) 97.9924%
- (c) 98.9924%
- (d) 99.9924%

Ans. (d) : 365 days have 525600 minutes. It takes 2 minutes to restart the computer in 2017. Computer crash for 20 times.

i.e system was restar – $20 \times 2 = 40$ minutes

So, $525600 - 40 = 525560$ available.

System was available for = 525560 minutes software

$$\text{Availability} = \frac{.525560}{525600} = .999924 \text{ or } = 99.9924\%$$

15. Match the 5 CMM Maturity levels/CMMI staged representations in List-I with their characterizations in List-II :

List - I		List - II	
a. Initial	i.	Processes are improved quantitatively and continually.	
b. Repeatable	ii.	The plan for a project comes from a template for plans.	
c. Defined	iii.	The plan uses processes that can be measured quantitatively.	
d. Managed	iv.	There may not exist a plan or it may be abandoned.	
e. Optimizing	v.	There's a plan and people stick to it.	

Code :

	(a)	(b)	(c)	(d)	(e)
(a)	(iv)	(v)	(i)	(iii)	(ii)
(b)	(i)	(ii)	(iv)	(v)	(iii)
(c)	(v)	(iv)	(ii)	(iii)	(i)
(d)	(iv)	(v)	(ii)	(iii)	(i)

Ans. (d) :

16. Coupling is a measure of the strength of the interconnections between software modules. Which of the following are correct statements with respect to module coupling ?

P : Common coupling occurs when one module controls the flow of another module by passing it information on what to do.

Q : In data coupling, the complete data structure is passed from one module to another through parameters.

R : Stamp coupling occurs when modules share a composite data structure and use only parts of it.

Code :

(a) P and Q only	(b) P and R only
(c) Q and R only	(d) All of P, Q and R

Ans. (c) : According to Myer's classification, which defines seven levels of coupling, from tightest to loosest.

- **Content Coupling**– Modules rely on each other's internal data or internal organization.
- **Common Coupling**– Modules shares the same global data.
- **External Coupling**– Modules share an externally imposed data format, communication protocol or device interface.

- **Control Coupling**– One module controls the flow of another, such as by passing it a flag or other information.
- **Stamp Coupling**–Modules share a composite data structure but use different parts of it.
- **Data Coupling**– Modules share data through parameters such as in a subroutine call.
- **Message Coupling**– Modules communicate by passing messages.

17. A software design pattern often used to restrict access to an object is :

- (a) adapter
- (b) decorator
- (c) delegation
- (d) proxy

Ans. (d) : A proxy is a wrapper or agent object that is being called by the client to access the real serving object behind the scenes. Use of the proxy can simply be forwarding to the real object or can provide additional logic.

Adapter– Match interface of different classes.

Decorator– Add responsibilities objects dynamically. Effectively in the instantiation process, object-creation patterns use delegation effectively to get the job done.

18. Reasons to re-engineer a software include :

P : Allow legacy software to quickly adapt to the changing requirements

Q : Upgrade to newer technologies / platforms / paradigm (for example, object-oriented)

R : Improve software maintainability

S : Allow change in the functionality and architecture of the software

Code :

- (a) P, R and S only
- (b) P and R only
- (c) P, Q and S only
- (d) P, Q and R only

Ans. (c) : Allow legacy software to quickly adapt to the changing requirements. **True**

Upgrade to newer technologies / platforms / paradigm (for example, object-oriented). **True**

Improve software maintainability. **False**

Allow change in the functionality and architecture of the software. **True**

19. Which of the following is not a key strategy followed by the clean room approach to software development ?

- (a) Formal specification
- (b) Dynamic verification
- (c) Incremental development
- (d) Statistical testing of the system

Ans. (b) : The clean room approach to software development is based on five key strategies :

1. Formal specification.
2. Incremental development.
3. Structured programming only.
4. Static verification.
5. Static testing of the system.

But dynamic verification is not the clean room approach for software development, so option (b) is correct.

20. Which of the following statements is/are True ?

P: Refactoring is the process of changing a software system in such a way that it does not alter the external behavior of the code yet improves the internal architecture.

Q: An example of refactoring is adding new features to satisfy a customer requirement discovered after a project is shipped.

Code :

- (a) P only (b) Q only
(c) Both P and Q (d) Neither P nor Q

Ans. (a) : Refactoring is the process of changing a software system in such a way that it does not alter the external behavior of the code yet improves the internal architecture. **True**

An example of refactoring is adding new features to satisfy a customer requirement discovered after a project is shipped. **False**

21. The solution of the recurrence relation

$T(m) = T(3m/4) + 1$ is :

- (a) θ (lg m) (b) θ (m)
- (c) θ (mlg m) (d) θ (lglg m)

Ans. (a) : $T(m) = aT(m/b) + f(m)$

Compare both equation $a = 1$; $b = 4/3$

$$m^{\log_b a} = m^0 = 1$$

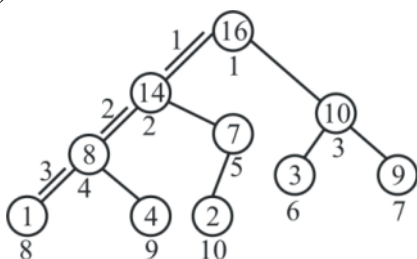
Whenever $m^{\log_b a}$ and $f(m)$ is equal then answer is $f(m) \cdot \log m$

Hence $1 * \log(m) = \log(m)$

22. Consider the array $A = \langle 4, 1, 3, 2, 16, 9, 10, 14, 8, 7 \rangle$. After building heap from the array A , the depth of the heap and the right child of max-heap are _____ and _____ respectively. (Root is at level 0).

- (a) 3, 14 (b) 3, 10
(c) 4, 14 (d) 4, 10

Ans. (b) :



This is max heap and right of root is 10.

The depth is no. of edge from root to given node so depth is 3.

23. A hash function h defined $h(\text{key}) = \text{key} \bmod 7$, with linear probing, is used to insert the keys 44, 45, 79, 55, 91, 18, 63 into a table indexed from 0 to 6. What will be the location of key 18?

- (a) 3 (b) 4
(c) 5 (d) 6

Ans. (c) :

0	1	2	3	4	5	6

Keys = 44, 45, 79, 55, 91, 18, 63, insert one by one,

start with 44, $44\%7 = 2$

0	1	2	3	4	5	6
		44				

Insert 41, $45\%7 = 3$

0	1	2	3	4	5	6
		44	45			

Insert 79,

0	1	2	3	4	5	6
		44	45	79		

$79\%7 = 27$ collision

→ check

Next cell; collision ⇒ check next cell ⇒ fit at 4

Insert – 55, $55\%7 = 6$

0	1	2	3	4	5	6
		44	45	79		55

Insert 18, $18\%7 = 4$ collision ⇒ check next cell ⇒ fit at 5

0	1	2	3	4	5	6
91		44	45	79	18	55

Therefore 18 fits at location 5.

24. Which of the following algorithms solves the single-source shortest paths ?

- Prim's algorithm
- Floyd - Warshall algorithm
- Johnson's algorithm
- Dijkstra's algorithm

Ans. (d) : Dijkstra's algorithm to find the shortest path from a single source vertex to all other vertices.

- Create a set spt set (shortest path tree set).
- Assign Q distance value to all vertices in the input graph.
- While spt set don't include all vertices.

25. A text is made up of the characters A, B, C, D, E each occurring with the probability 0.08, 0.40, 0.25, 0.15 and 0.12 respectively. The optimal coding technique will have the average length of :

- 2.4
- 1.87
- 3.0
- 2.15

Ans. (d) : A = 0.08, B = 0.40, C = .25, D=.15, E=0.12

A = 0.08

B = 0.40

C = 0.25

D = 0.15

E = 0.12

Code 1 : $(0.08) \times 3 + 0.40 \times 3 + 0.15 \times 3; 8 \times 3$
 $= 0.25 \times 3 = 3$

Code 2 : $= 0.12 \times 3 + 0.4 \times 2 + 0.15 \times 2 + 0.08 \times 3 + 0.25 \times (2)$
 $= 2.22$

26. A binary search tree in which every non-leaf node has non-empty left and right subtrees is called a strictly binary tree. Such a tree with 19 leaves :
- (a) cannot have more than 37 nodes
 - (b) has exactly 37 nodes
 - (c) has exactly 35 nodes
 - (d) cannot have more than 35 nodes

Ans. (b) : $2(n) - 1$, where n is the for the leaves nodes. so by that way we have exactly 37 nodes.

$$2(19) - 1$$

$$= 38 - 1 = 37$$

27. Match the following with respect to algorithm paradigms :

List - I	List - II
a. The 8-Queen's problem	i. Dynamic programming
b. Single-Source shortest paths	ii. Divide and conquer
c. STRASSEN's Matrix multiplication	iii. Greedy approach
d. Optimal binary search trees	iv. Backtracking

Code :

	(a)	(b)	(c)	(d)
(a)	(iv)	(i)	(iii)	(ii)
(b)	(iv)	(iii)	(i)	(ii)
(c)	(iii)	(iv)	(ii)	(i)
(d)	(iv)	(iii)	(ii)	(i)

Ans. (d) :

28. The maximum number of comparisons needed to sort 9 items using radix sort is (assume each item is 5 digit octal number) :
- (a) 45
 - (b) 72
 - (c) 360
 - (d) 450

Ans. (c) : Number of items = 9

Each item is assumed to be octal number of 5 digit.

Base of octal number system = 8

Maximum number of comparisons required to sort these 9 items.

= number of items $\times$ number of digits $\times$ base of number system

= $9 \times 5 \times 8 = 360$

29. A 5-ary tree is tree in which every internal node has exactly 5 children. The number of left nodes in such a tree with 8 internal nodes will be :
- (a) 30
 - (b) 33
 - (c) 45
 - (d) 125

Ans. (b) : In complete n -ary tree,

$$L = (n - 1) * I + 1$$

Where L = No. of leaves, I = no. of internal nodes

Here $n = 5, I = 8$

Therefore, $L = (5 - 1) * 8 + 1 = 4 * 8 + 1 = 33$

30. Consider a Boolean function of 'n' variables. The order of an algorithm that determines whether the Boolean function produces a output 1 is :

- (a) Logarithmic (b) Linear
(c) Quadratic (d) Exponential

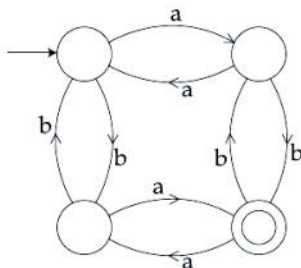
Ans. (d) : N boolean variable $\Rightarrow 2^n$ rows in truth table. For saying o/p 1-for the given function in worst case it needs to check every possible row $\Rightarrow O(2^n) \Rightarrow$ Exponential.

31. Two finite state machines are said to be equivalent if they :

- (a) Have the same number of edges
(b) Have the same number of states
(c) Recognize the same set of tokens
(d) Have the same number of states and edges

Ans. (c) : Two finite state machines are said to be equivalent if they, Recognize the same set of tokens.

32. The finite state machine given in figure below recognizes :



- (a) any string of odd number of a's
(b) any string of odd number of b's
(c) any string of even number of a's and odd number of b's
(d) any string of odd number of a's and odd number of b's

Ans. (d) :

- (a) Any string of odd number of a's – a, aaa these are not accepted.
(b) Any string of odd number of b's– b, bbb these are not accepted.
(c) Any string of even number of a's and odd number of b's– aab, aabbb, aaaabbb, aabb, These are not accepted.
(d) Any string of odd number of a's and odd number of b's– ab, ba, aaab, bbba, abbb, aaabbb These are accepted.

33. A pushdown automata behaves like a Turing machine when the number of auxiliary memory is :

- (a) 0 (b) 1
(c) 1 or more (d) 2 or more

Ans. (d) : Pushdown automata have only one stack with 2 stacks we can implement queue $\Rightarrow T(n+2)$ is sufficient form Turning Machine.

Moreover expressive power of $T(n + 2) = T(n + 3) = T(n + 4)$ etc.

Therefore we need one more extra auxiliary stack memory for PDA or we need at least 2 auxiliary memory to implement TM

Option (d) is correct.

34. Pushdown automata can recognize language generated by_____.

- (a) Only context free grammar
- (b) Only regular grammar
- (c) Context free grammar or regular grammar
- (d) Only context sensitive grammar

Ans. (c) : Pushdown automata recognize regular as well as context free grammar.

35. To obtain a string of n Terminals from a given Chomsky normal form grammar, the number of productions to be used is :

- (a) $2n - 1$
- (b) $2n$
- (c) $n + 1$
- (d) n^2

Ans. (a) : In CNF, the height of the syntax tree is $2n - 1$, i.e. –

$2n - 1$ productions are used to obtain a string on n terminals.

36. Consider the following two Grammars :

$G_1 : S \rightarrow SbS \mid a$

$G_2 : S \rightarrow aB \mid ab, A \rightarrow GAB \mid a, B \rightarrow ABb \mid b$

Which of the following option is correct ?

- (a) Only G_1 is ambiguous
- (b) Only G_2 is ambiguous
- (c) Both G_1 and G_2 are ambiguous
- (d) Both G_1 and G_2 are not ambiguous

Ans. (c) : For grammar G_1

try to make string aba then we will get two parse tree,

So, G_1 is ambiguous

for grammar G_2

try to make string ab then we will get two

parse tree

So, G_2 is ambiguous.

37. Context sensitive language can be recognized by a :

- (a) Finite state machine
- (b) Deterministic finite automata
- (c) Non-deterministic finite automata
- (d) Linear bounded automata

Ans. (d) : Context sensitive language can be recognized by a linear bounded automata for context sensitive grammar we need linear bounded automata.

38. The set $A = \{ 0^n 1^n 2^n \mid n = 1, 2, 3, \dots \}$ is an example of a grammar that is :

- (a) Context sensitive
- (b) Context free
- (c) Regular
- (d) None of the above

Ans. (a) : $a^n b^n c^n$ is context sensitive grammar. We need two stack.

39. A bottom-up parser generates :

- (a) Left-most derivation in reverse
- (b) Right-most derivation in reverse
- (c) Left-most derivation
- (d) Right-most derivation

Ans. (b) : A bottom-up parser generates is right-most derivation in reverse.

40. Consider the following statements() :

S₁ : There exists no algorithm for deciding if any two Turing machines M1 and M2 accept the same language.

S₂ : The problem of determining whether a Turing machine halts on any input is undecidable.

Which of the following options is correct ?

- (a) Both S₁ and S₂ are correct
- (b) Both S₁ and S₂ are not correct
- (c) Only S₁ is correct
- (d) Only S₂ is correct

Ans. (a) :

S₁ : Equivalence test of two TM is undecidable.

S₂ : Halting problem of TM is undecidable.

41. A slotted ALOHA network transmits 200-bit frames using a shared channel with a 200 Kbps bandwidth. Find the throughput of the system, if the system (all stations put together) produces 250 frames per second :

- (a) 49
- (b) 368
- (c) 149
- (d) 151

Ans. (a) : The frame transmission time = $200/200\text{Kbps}$
= 1ms

$$G = 1/4$$

So, $S = Ge^{-G} = 0.195 = 19.5\%$

This means that the throughput is $250 \times 0.195 = 49$ frames only 49 out of 250 frames will survive.

42. The period of a signal is 100 ms. Its frequency is _____.

- (a) 100^3 Hertz
- (b) 10^{-2} KHz
- (c) 10^{-3} KHz
- (d) 10^5 Hertz

Ans. (b) : Frequency is the inverse of time period.

$$f = 1/t = 1/100 \text{ ms} = \frac{1}{10^{-1}} \text{ s}$$
$$= 10\text{Hz} = 10^{-2} \text{ KHz}$$

43. The dotted-decimal notation of the following IPV4 address in binary notation is _____.

10000001 00001011 00001011 11101111

- (a) 111.56.45.239
- (b) 129.11.10.238
- (c) 129.11.11.239
- (d) 111.56.11.239

Ans. (c) : $10000001 = 2^7 + 2^0 = 129$

$$00001011 = 2^0 + 2^3 + 2^1 = 11$$

$$11101111 = 2^0 + 2^1 + 2^2 + 2^3 + 2^5 + 2^6 + 2^7$$
$$= 239$$

Hence in dotted decimal notation, it would be 129.11.11.239.

- 44. Which of the following statements are true ?**
- (a) Advanced Mobile Phone System (AMPS) is a second generation cellular phone system.
 - (b) IS - 95 is a second generation cellular phone system based on CDMA and DSSS.
 - (c) The Third generation cellular phone system will provide universal personnel communication.

Code :

(a) (a) and (b) only (b) (b) and (c) only

(c) (a), (b) and (c) (d) (a) and (c) only

Ans. (b) :

(a) Its false, advance mobile phone system (AMPS) is first generation cellular phone system. It uses separate frequencies or channels for conversation that's why it requires considerable bandwidth for large number of users.

(b) Its true, IS-95 is a second generation cellular phone system based on CDMA and DSSS.

(c) Its true, the third generation cellular phone system will provide universal personnel communication.

- 45. Match the following symmetric block ciphers with corresponding block and key sizes :**

List - I		List - II	
a. DES	i.	block size 64 and key size ranges between 32 and 448	
b. IDEA	ii.	block size 64 and key size 64	
c. BLOW FISH	iii.	block size 128 and key sizes 128, 192, 256	
d. AES	iv.	block size 64 and key size 128	

Code :

	(a)	(b)	(c)	(d)
(a)	(iv)	(ii)	(i)	(iii)
(b)	(ii)	(iv)	(i)	(iii)
(c)	(ii)	(iv)	(iii)	(i)
(d)	(iv)	(ii)	(iii)	(i)

Ans. (b) :

- 46. Which of the following statements are true ?**
- (a) Three broad categories of Networks are
 - (i) Circuit Switched Networks
 - (ii) Packet Switched Networks
 - (iii) Message Switched Networks
 - (b) Circuit Switched Network resources need not be reserved during the set up phase.
 - (c) In packet switching there is no resource allocation for packets.

Code :

(a) (a) and (b) only (b) (b) and (c) only

(c) (a) and (c) only (d) (a), (b) and (c)

Ans. (c): In circuit switching a dedicated. Resources are allocated. e.g. PSTN telephone uses circuit switching. In packet switching, No. resources are allocated.

- 47. In Challenge-Response authentication the claimant**
- (a) Proves that she knows the secret without revealing it
 - (b) Proves that she doesn't know the secret
 - (c) Reveals the secret
 - (d) Gives a challenge

Ans. (a) : Challenge– Response authentication is a computer security protocol in which one party ask a question and another party must provide a valid answer. i.e. she knows the secret without revealing it or will provide the secret when it is asked.

- 48. Decrypt the message “WTAAD” using the Caesar Cipher with key=15.**

- (a) LIPPS
- (b) HELLO
- (c) OLLEH
- (d) DAATW

Ans. (b) : In Caesar Cipher algorithm encryption and decryption takes place. During encryption right shift takes place depending upon the key and in decryption left shift takes place depending on key.

In the given question key value is 15 and decryption asked–

Given message is WTAAD

Left shift 15 on W is H

Left shift 15 on T is E

Left shift 15 on A is L

Left shift 15 on L is L

Left shift 15 on D is O

The decrypted message will HELLO.

- 49. To guarantee correction of upto t errors, the minimum Hamming distance $d_{\min}$ in a block code must be _____.**

- (a) $t + 1$
- (b) $t - 2$
- (c) $2t - 1$
- (d) $2t + 1$

Ans. (d) : To guarantee correction of up to t error in all cases, the minimum flaming distance in a block code must be $d_{\min} = 2t + 1$.

- 50. Encrypt the Message “HELLO MY DEARZ” using Transposition Cipher with**

Key $\begin{cases} \text{Plain Text 2413} \\ \text{Cipher Text 1234} \end{cases}$

- (a) HLLEO YM AEDRZ
- (b) EHOLL ZYM RAED
- (c) ELHL MDOY AZER
- (d) ELHL DOMY ZAER

Ans. (c):Arranging the message in tabular format, we get,

1	H	O	E
2	E	M	A
3	L	Y	R
4	L	D	Z

Now applying the key and writing out the corresponding letters from plain text according to key 2413 we get ELHL MDOY AZER.

- 51. At a particular time of computation, the value of a counting semaphore is 10. Then 12 P operations and “x” V operations were performed on this semaphore. If the final value of semaphore is 7, x will be :**

(a) 8 (b) 9
(c) 10 (d) 11

Ans. (b) : Current value of CS = 10

$$CS - 12 + X = 7$$

$$X = 7 + 12 - 10 = 9$$

- 52. In a paged memory, the page hit ratio is 0.40. The time required to access a page in secondary memory is equal to 120 ns. The time required to access a page in primary memory is 15 ns. The average time required to access a page is _____.**

(a) 105 (b) 68
(c) 75 (d) 78

Ans. (d) : $\mu_T = 0.40$

$$T_s = 120 \text{ ns}$$

$$T_m = 15 \text{ ns}$$

$$\begin{aligned} T_{\text{avg}} &= 0.40 \times 15 + 0.60 (15 + 120) \text{ ns} \\ &= 6.00 + .60 \times 135 \\ &= 6 + 8.10 \\ &= 6.8 \end{aligned}$$

- 53. In a multi-user operating system, 30 requests are made to use a particular resource per hour, on an average. The probability that no requests are made in 40 minutes, when arrival pattern is a poisson distribution, is _____.**

(a) e^{-15} (b) $1 - e^{-15}$
(c) $1 - e^{-20}$ (d) e^{-20}

Ans. (d) : In hour – 30 Requests sent

i.e. in 60 minutes – 30 ,, ,,

$$\text{in 40 minutes} = \frac{30 \times 40}{60} = 20 \text{ Request sent}$$

$\lambda = 20$ [λ is number of requests that could is sent in 40 minutes]

Actual no. of request sent in 40 minutes = 0 ie = $x = 0$

As arrival pattern is a poison distribution

$$\text{Probability } (x = 0) = \frac{e^{-\lambda} \lambda^x}{x!} \Rightarrow \frac{e^{-20} + 20^0}{0!} = e^{-20}$$

$$D = e^{-20}$$

- 54. Normally user programs are prevented from handling I/O directly by I/O instructions in them. For CPUs having explicit I/O instructions, such I/O protection is ensured by having the I/O instructions privileged. In a CPU with memory mapped I/O, there is no explicit I/O instruction. Which one of the**

following is true for a CPU with memory mapped I/O ?

- (a) I/O protection is ensured by operating system routines.
- (b) I/O protection is ensured by a hardware trap.
- (c) I/O protection is ensured during system configuration.
- (d) I/O protection is not possible.

Ans. (a) : User application are not allowed to perform I/O in user mode – All I/O requests are handled through system calls that must be performed in kernel mode.

I/O protection is ensured by operating system routines.

55. Which UNIX/Linux command is used to make all files and sub-directories in the directory “progs” executable by all users ?

- (a) `chmod - R a + x progs`
- (b) `chmod - R 222 progs`
- (c) `chmod - X a + x progs`
- (d) `chmod - X 222 progs`

Ans. (a) : "`chmod - R a + x progs`" command is used to make all files and sub-directories in the directory “progs” executable by all users in the Linux.

56. Which of the following statements are true ?

- (a) External Fragmentation exists when there is enough total memory space to satisfy a request but the available space is contiguous.
- (b) Memory Fragmentation can be internal as well as external.
- (c) One solution to external Fragmentation is compaction.

Code :

- (a) (a) and (b) only (b) (a) and (c) only
- (c) (b) and (c) only (d) (a), (b) and (c)

Ans. (c) : External fragmentation exists when there is enough total memory space to satisfy a request but the available space is contiguous is fail.

External fragmentation exists when there is enough total memory space to satisfy a request but travel spell available space is continuous in so this is false the definition of external fragmentation is that whenever you have canting venom or any process want some data for some space from the main memory space is available in the main memory but it is not contiguous ok for example let us.

57. Page information in memory is also called as Page Table. The essential contents in each entry of a page table is/are _____.

- (a) Page Access information
- (b) Virtual Page number
- (c) Page Frame number
- (d) Both virtual page number and Page Frame Number

Ans. (c) : Page information in memory is also called as Page Table. The essential contents in each entry of a page table is Page Frame number.

58. Consider a virtual page reference string 1, 2, 3, 2, 4, 2, 5, 2, 3, 4. Suppose LRU page replacement algorithm is implemented with 3 page frames in main memory. Then the number of page faults are_____.

- (a) 5 (b) 7
(c) 9 (d) 10

Ans. (b) : For the 1st 3 pages, there would be page faults.

Next 2 is already in frame Next least recently used 1 is replaced by 4.

Next 2 is already in frame. Then LRU 3 is replaced by 5.

Then 2 is already in frame. Then 4 is replaced by 3 and 5 is replaced by 4. Hence a total of 7 page faults.

3	3	5	5	4
2	2	2	2	2
1	4	4	3	3

$$3 + 1 + 1 + 1 + 1 = 7$$

59. Consider the following three processes with the arrival time and CPU burst time given in milliseconds :

Process	Arrival Time	Burst Time
P ₁	0	7
P ₂	1	4
P ₃	2	8

The Gantt Chart for preemptive SJF scheduling algorithm is_____.

- (a)

P ₁	P ₂	P ₃
----------------	----------------	----------------

0 7 13 21
- (b)

P ₁	P ₂	P ₁	P ₃
----------------	----------------	----------------	----------------

0 1 5 11 19
- (c)

P ₁	P ₂	P ₃
----------------	----------------	----------------

0 7 11 19
- (d)

P ₂	P ₃	P ₁
----------------	----------------	----------------

0 4 12 19

Ans. (b) : Pre Emptive shortest remaining time first scheduling, i.e. that process will be scheduled on the CPU which will be having least remaining burst time. (required time at the CPU).

The processes are scheduled and executed as given in the below Gantt chart.

P ₁	P ₂	P ₁	P ₃
----------------	----------------	----------------	----------------

0 1 5 11 19

60. In which of the following scheduling criteria, context switching will never take place ?

- (a) ROUND ROBIN
(b) Preemptive SJF
(c) Non-preemptive SJF
(d) Preemptive priority

Ans. (c) : Shortest job first or shortest job next, is a scheduling policy that selects the waiting process with the smallest execution time to execute next SJF is non-preemptive algorithm.

Except Non-Preemptive SJF, all are preemptive. CPU will have leave the shortest process only after its full execution, once process is started to execute.

61. In RDBMS, which type of Join returns all rows that satisfy the join condition ?

- (a) Inner Join (b) Outer Join
(c) Semi Join (d) Anti Join

Ans. (a) : An Inner join (Sometimes called a simple join) is a join of two or more tables that return only those rows that satisfy the join condition.

Outer Joins— An outer join extends the result of a simple join. An outer join returns all rows that satisfy the join condition and also return some or all of those rows from one table for which no rows from the other satisfy the join condition.

62. Consider a relation book (title, price) which contains the titles and prices of different books. Assuming that no two books have the same price, what does the following SQL query list ?

Select title
from book as B
where (select count (*)
from book as T
where T.price > B.price) < 7

- (a) Titles of the six most expensive books.
(b) Title of the sixth most expensive books.
(c) Titles of the seven most expensive books.
(d) Title of the seventh most expensive books.

Ans. (c) : Select title
from book as B
where (select count (*)
from book as T
where T.price > B.price) < 7
Titles of the seven most expensive books.

63. In a Hierarchical database, a hashing function is used to locate the _____.

- (a) Collision (b) Root
(c) Foreign Key (d) Records

Ans. (b) : In a Hierarchical database, a hashing function is used to locate the Root.

64. Relations produced from E - R Model will always be in _____.

- (a) 1 NF (b) 2 NF
(c) 3 NF (d) 4 NF

Ans. (a) : Simple atomic values non multivalued attributes. Generally ER diagram is design level task and relational model is implementation. Whenever relational model is implemented from ER diagram. We get rid of multi valued attribute.

Multi valued attributes are removed as M : N binary relation by repeating becomes 1NF. We cannot guarantee any higher normalization.

65. Consider the following schedules involving two transactions.

$S_1 : r_1(X) ; r_1(Y) ; r_2(X) ; r_2(Y) ; w_2(Y) ; w_1(X)$
 $S_2 : r_1(X) ; r_2(X) ; r_2(Y) ; w_2(Y) ; r_1(Y) ; w_1(X)$

Which one of the following statements is correct with respect to above ?

- (a) Both S_1 and S_2 are conflict serializable.
- (b) Both S_1 and S_2 are not conflict serializable.
- (c) S_1 is conflict serializable and S_2 is not conflict serializable.
- (d) S_1 is not conflict serializable and S_2 is conflict serializable.

Ans. (d) :

S_1

Cycle

S_2

No cycle

Conflict operations are : RW, WR, WW
 Non-conflict operation : RR if there is cycle found then it can't be conflict serializable.
 In S_1 we found a cycle so its not conflict serializable but in S_2 there's no cycle found so its conflict serializable.

66. For a database relation $R(a, b, c, d)$ where the domains of a, b, c and d include only atomic values, and only the following functional dependencies and those that can be inferred from them hold :

- $a \rightarrow c$
- $b \rightarrow d$

The relation is in _____.

- (a) First normal form but not in second normal form
- (b) Second normal form but not in third normal form
- (c) Third normal form
- (d) BCNF

Ans. (a) : We can not inferred extra dependencies and transitive dependencies so key for $a \rightarrow c$ & $b \rightarrow d$ relation will be- ab
 $ab^+ = \{a, b, c, d\}$, So, ab will be the key for this relation.
 So, BCNF is says left hand side should be super key.
 So, option (d) is wrong.

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Second NF says that there should be not any partial dependencies i.e. $PA \rightarrow \text{Non-key}$.

Drive attributes $\rightarrow$ Non-key.

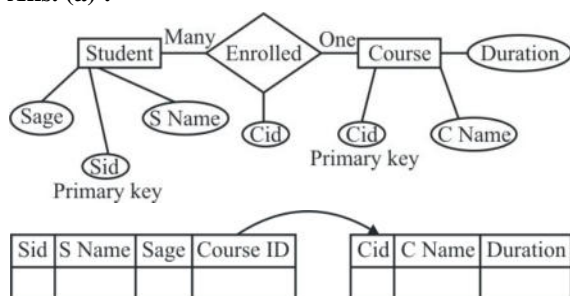
So, option (b) is also wrong.

So, option (a) is correct.

67. A many-to-one relationship exists between entity sets r_1 and r_2 . How will it be represented using functional dependencies if $Pk(r)$ denotes the primary key attribute of relation r ?

- (a) $Pk(r_1) \rightarrow Pk(r_2)$
- (b) $Pk(r_2) \rightarrow Pk(r_1)$
- (c) $Pk(r_2) \rightarrow Pk(r_1)$ and $Pk(r_1) \rightarrow Pk(r_2)$
- (d) $Pk(r_2) \rightarrow Pk(r_1)$ or $Pk(r_1) \rightarrow Pk(r_2)$

Ans. (a) :



From the Sid we can find out the CID so, option (1) is correct.

68. Database systems that store each relation in a separate operating system file may use the operating system's authorization scheme, instead of defining a special scheme themselves. In this case, which of the following is false ?

- (a) The administrator enjoys more control on the grant option.
- (b) It is difficult to differentiate among the update, delete and insert authorizations.
- (c) Cannot store more than one relation in a file.
- (d) Operations on the database are speeded up as the authorization procedure is carried out at the operating system level.

Ans. (d) : Database systems that store each relation in a separate operating system file may use the operating system's authorization scheme, instead of defining a special scheme themselves.

- (a) The administrator enjoys more control on the grant option.
- (b) It is difficult to differentiate among the update, delete and insert authorizations.
- (c) Cannot store more than one relation in a file.

69. Let $R_1(a, b, c)$ and $R_2(x, y, z)$ be two relations in which a is the foreign key of R_1 that refers to the primary key of R_2 . Consider following four options.

- (a) Insert into R_1
- (b) Insert into R_2
- (c) Delete from R_1
- (d) Delete from R_2

Which of the following is correct about the referential integrity constraint with respect to above ?

- (a) Operations (a) and (b) will cause violation.
- (b) Operations (b) and (c) will cause violation.
- (c) Operations (c) and (d) will cause violation.
- (d) Operations (d) and (a) will cause violation.

Ans. (d) :

R			R to PR of R		
a(PR)	b	c	x d(fR) referential	y	z
1			2		
2			1		

Insert into R cannot cause any violation.

Insert into S can cause violation if any value is inserted into d of S, which value is not in a of R. Delete from S would cause no. violation.

Delete from R would cause violation if it any tuple is deleted, and as a result a value in a gets deleted which is referred to by d in S.

70. Consider a hash table of size seven, with starting index zero, and a hash function $(7x+3) \bmod 4$. Assuming the hash table is initially empty, which of the following is the contents of the table when the sequence 1, 3, 8, 10 is inserted into the table using closed hashing ? Here “_” denotes an empty location in the table.

- (a) 3, 10, 1, 8, _ , _ , _
- (b) 1, 3, 8, 10, _ , _ , _
- (c) 1, _ , 3, _ , 8, _ , 10
- (d) 3, 10, _ , _ , 8, _ , _

Ans. (a) :

3	10	1	8			
0	1	2	3	4	5	6

$$h(x) = (7x + 3) \bmod 4$$

$$h(1) = (7 \times 1 + 3) \bmod 4 \Rightarrow 2$$

$$h(2) = (7 \times 3 + 3) \bmod 4 \Rightarrow 0$$

$$h(8) = (7 \times 8 + 3) \bmod 4 \Rightarrow 3$$

$$h(10) = (7 \times 10 + 3) \bmod 4 \Rightarrow 1$$

So, here 3 empty options. So, option (a) is correct.

71. In Artificial Intelligence (AI), an environment is uncertain if it is _____.

- (a) Not fully observable and not deterministic
- (b) Not fully observable or not deterministic
- (c) Fully observable but not deterministic
- (d) Not fully observable but deterministic

Ans. (b) : We say an environment is uncertain if it is not fully observable or not deterministic option (b).

72. In Artificial Intelligence (AI), a simple reflex agent selects actions on the basis of_____.

- (a) current percept, completely ignoring rest of the percept history.
- (b) rest of the percept history, completely ignoring current percept.
- (c) both current percept and complete percept history.
- (d) both current percept and just previous percept.

Ans. (a) : In Artificial Intelligence (AI), a simple reflex agent selects actions on the basis of current percept, completely ignoring rest of the percept history.

73. In heuristic search algorithms in Artificial Intelligence (AI), if a collection of admissible heuristics $h_1, \dots, h_m$ is available for a problem and none of them dominates any of the others, which should we choose?

- (a) $h(n) = \max\{h_1(n), \dots, h_m(n)\}$
- (b) $h(n) = \min\{h_1(n), \dots, h_m(n)\}$
- (c) $h(n) = \text{avg}\{h_1(n), \dots, h_m(n)\}$
- (d) $h(n) = \text{sum}\{h_1(n), \dots, h_m(n)\}$

Ans. (a) : In heuristic search algorithms in Artificial Intelligence (AI), if a collection of admissible heuristics $h_1, \dots, h_m$ is available for a problem and none of them dominates any of the others then $h(n) = \max\{h_1(n), \dots, h_m(n)\}$.

74. Consider following sentences regarding A*, an informed search strategy in Artificial Intelligence (AI).

- (a) A* expands all nodes with $f(n) < C^*$.
- (b) A* expands no nodes with $f(n) / C^*$.
- (c) Pruning is integral to A*.

Here, C^* is the cost of the optimal solution path.

Which of the following is correct with respect to the above statements ?

- (a) Both statement (a) and statement (b) are true.
- (b) Both statement (a) and statement (c) are true.
- (c) Both statement (b) and statement (c) are true.
- (d) All the statements (a), (b) and (c) are true.

Ans. (a) : A* expands all nodes with $f(n) < C^*$. **True**
A* expands no nodes with $f(n) / C^*$. **True**
Pruning is integral to A*. **False**

75. Consider a vocabulary with only four propositions A, B, C and D. How many models are there for the following sentence ?

$B \vee C$

- (a) 10
- (b) 12
- (c) 15
- (d) 16

Ans. (b) :

A	B	C	D	$B \vee C$
F	F	F	F	F
F	F	F	T	F
F	F	T	F	T
F	F	T	T	T
F	T	F	F	T
F	T	F	T	T
F	T	T	F	T
F	T	T	T	T
T	F	F	F	F
T	F	F	T	F
T	F	T	F	T
T	F	T	T	T
T	T	F	F	T
T	T	F	T	T
T	T	T	F	T
T	T	T	T	T

So, from the given sentence false (F) occurs only if B and C are false (F) which occurs 4 times.

So, $16 - 4 = 12$

So, option (b) is write.

76. Consider the following statements :

(a) False $\models$ True

(b) If $\alpha \models (\beta \wedge \gamma)$ then $\alpha \models \beta$ and $\alpha \models \gamma$.

Which of the following is correct with respect to the above statements ?

(a) Both statement (a) and statement (b) are false.

(b) Statement (a) is true but statement (b) is false.

(c) Statement (a) is false but statement (b) is true.

(d) Both statement (a) and statement (b) are true.

Ans. (d) : (a) False $\models$ True

Suppose variable is P

$$\sim P \rightarrow P \quad [p + q = \sim p \vee q]$$

$$\sim p \vee p$$

$$= p \quad [\because 0 + 1 = 1]$$

(b) If $\alpha \models (\beta \wedge \gamma)$ then $\alpha \models \beta$ and $\alpha \models \gamma$

$$\alpha \models (\beta \wedge \gamma)$$

$$\sim \alpha \rightarrow (\beta \wedge \gamma) \quad [p \rightarrow q = \sim p \vee q]$$

$$\sim \alpha \vee (\beta \wedge \gamma)$$

$$\sim \alpha \vee \beta \wedge \sim \alpha \vee \gamma$$

[Which are equivalent to $\sim p \vee q$ so we can write it as $p \rightarrow q$]

$$\alpha \rightarrow \beta \wedge \alpha \rightarrow \gamma$$

$$\alpha \models \beta \text{ and } \alpha \models \gamma$$

So, both statement are true.

77. Consider the following English sentence :

“Agra and Gwalior are both in India”.

A student has written a logical sentence for the above English sentence in First-Order Logic using predicate In(x, y), which means x is in y, as follows :

In (Agra, India) $\vee$ In(Gwalior, India)

Which one of the following is correct with respect to the above logical sentence ?

- (a) It is syntactically valid but does not express the meaning of the English sentence.
- (b) It is syntactically valid and expresses the meaning of the English sentence also.
- (c) It is syntactically invalid but expresses the meaning of the English sentence.
- (d) It is syntactically invalid and does not express the meaning of the English sentence.

Ans. (a) : Given that "Agra and Gwalior are both in India"

Representation– In (x, y) means x is in India

In (Agra, India)(1)

In (Gwalior, India)(2)

According to the English statement, in (Agra, India) is should be true, In (Gwalior, India) is should be true there use conjunction as connector $\implies$ In (Agra, India) $\wedge$ In (Gwalior, India).

But given that In (Agra, India) $\vee$ In (Gwalior, India) $\implies$ syntactically correct but not represent the given. English sentence.

78. Consider the following two sentences :

- (a) The planning graph data structure can be used to give a better heuristic for a planning problem.
- (b) Dropping negative effects from every action schema in a planning problem results in a relaxed problem.

Which of the following is correct with respect to the above sentences ?

- (a) Both sentence (a) and sentence (b) are false.
- (b) Both sentence (a) and sentence (b) are true.
- (c) Sentence (a) is true but sentence (b) is false.
- (d) Sentence (a) is false but sentence (b) is true.

Ans. (b) : (a) The planning graph data structure can be used to give a better heuristic for a planning problem.

True

(b) Dropping negative effects from every action schema in a planning problem results in a relaxed problem. **True**

So, both sentence (a) and sentence (b) are true.

79. A knowledge base contains just one sentence, $\exists x \text{ AsHighAs } (x, \text{ Everest})$. Consider the following two sentences obtained after applying existential instantiation.

- (a) AsHighAs (Everest, Everest)
- (b) AsHighAs (Kilimanjaro, Everest)

Which of the following is correct with respect to the above sentences ?

- (a) Both sentence (a) and sentence (b) are sound conclusions.

- (b) Both sentence (a) and sentence (b) are unsound conclusions.
- (c) Sentence (a) is sound but sentence (b) is unsound.
- (d) Sentence (a) is unsound but sentence (b) is sound.

Ans. (c) : Existential instantiation is valid if we replace the variable with an arbitrary new constant. We can give the name of an existing object.

So, AsHighAs (Everest, Everest)

Sentence (a) is sound but sentence (b) is unsound.

80. Consider the set of all possible five-card poker hands dealt fairly from a standard deck of fifty-two cards. How many atomic events are there in the joint probability distribution ?

- (a) 2, 598, 960
- (b) 3, 468, 960
- (c) 3, 958, 590
- (d) 2, 645, 590

Ans. (a) : Joint probability distribution that gives the probability that each of $x, y, \dots$

falls in any particular range or discrete set of values specified for that variable.

$$P_b = {}^{52}C_5 \Rightarrow 2598960$$

81. E is the number of edges in the graph and f is maximum flow in the graph. When the capacities are integers, the runtime of Ford-Fulberson algorithm is bounded by :

- (a) $O(E \cdot f)$
- (b) $O(E^2 \cdot f)$
- (c) $O(E \cdot f^2)$
- (d) $O(E^2 \cdot f^2)$

Ans. (a) : The Basic Idea of the Ford– Fulkerson algorithm is to find any augmenting path in the graph and to add a constant amount of flow to each edge in that path. It repeats this process until it cannot find any more augmenting paths. At the end, flow is maximized. You may be tempted to think that the runtime is $O(vf)$ because each augmenting path has $O(V)$ edges and each of those edge can be $O(f)$ increases in its capacity in the worst case.

However, finding an augmenting path requires a depth–first search of the graph which takes $O(E)$ time. Since, we can do at most f iterations, and each iteration takes $O(E+V)$ time, the worst-case run time is $O((E+V)f)$ which is $O(Ef)$.

82. Which of the following statements is false about convex minimization problem ?

- (a) If a local minimum exists, then it is a global minimum
- (b) The set of all global minima is convex set
- (c) The set of all global minima is concave set
- (d) For each strictly convex function, if the function has a minimum, then the minimum is unique

Ans. (c) : The following statements are true about the convex minimization problem:

- If a local minimum exists, then it is a global minimum.
- The set of all (global) minima is convex.
- For each strictly convex function, if the function has a minimum, then the minimum is unique.

83. The following LPP

$$\text{Maximize } z = 100x_1 + 2x_2 + 5x_3$$

Subject to

$$14x_1 + x_2 - 6x_3 + 3x_4 = 7$$

$$32x_1 + x_2 - 12x_3 \leq 10$$

$$3x_1 - x_2 - x_3 \leq 0$$

$$x_1, x_2, x_3, x_4 \geq 0$$

has

- (a) Solution : $x_1 = 100, x_2 = 0, x_3 = 0$
- (b) Unbounded solution
- (c) No solution
- (d) Solution : $x_1 = 50, x_2 = 70, x_3 = 60$

Ans. (b) : $14x_1 + x_2 - 6x_3 + 3x_4 = 7$

$$32x_1 + x_2 - 12x_3 \leq 10$$

$$3x_1 - x_2 - x_3 \leq 0$$

According to this the unbounded feasible solution cannot be determined, instead there are infinite many solutions. It is not possible to solve the problem in which this situation occurs.

84. Digital data received from a sensor can fill up 0 to 32 buffers. Let the sample space be $S = \{0, 1, 2, \dots, 32\}$ where the sample j denote that j of the buffers are full and $p(i) = \frac{1}{561} (33 - i)$.

Let A denote the event that the even number of buffers are full. Then $p(A)$ is :

- (a) 0.515
- (b) 0.785
- (c) 0.758
- (d) 0.485

Ans. (a) : Given,

$$\text{Probability of } i^{\text{th}} \text{ buffer getting full } p(i) = \frac{1}{562} (33 - i)$$

Probability of all even number of buffers are full is $P(A)$

$$P(A) = \sum_{i=0, 2, 4, 6, 8, \dots, 32} p(i) = \sum_{i=0}^{16} p(2i)$$

$$P(A) = \frac{1}{562} (33 + 31 + 29 + 27 + \dots + 1)$$

$$P(A) = \frac{1}{562} \times 289 = 0.51423$$

85. The equivalence of

$\neg \exists x Q(x)$ is :

- (a) $\exists x \neg Q(x)$
- (b) $\forall x \neg Q(x)$
- (c) $\neg \exists x \neg Q(x)$
- (d) $\forall x Q(x)$

Ans. (b) : To negate $\exists x : Q(x)$, we must claim that $Q(x)$ fails to hold for any possible x . So again we flip the quantifier and then negate the predicate :

$$\forall x \neg Q(x) \text{ i.e. option (b).}$$

86. If $A_i = \{-i, \dots, -2, -1, 0, 1, 2, \dots, i\}$

then $\bigcup_{i=1}^{\infty} A_i$ is :

- (a) $\mathbb{Z}$ (b) $\mathbb{Q}$
(c) $\mathbb{R}$ (d) $\mathbb{C}$

Ans. (a) : Here (a) is the right ans $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ are set of rational, real and complex numbers respectively.

$\mathbb{Z}$ is the set of integers, i.e. positive, negative or zero is the set of rational number, i.e. represented by a fraction a/b with a belonging to $\mathbb{Z}$ and b belonging to $\mathbb{Z}^* \mathbb{R}$ is the set of real number, i.e. all numbers that can actually exist, it contains in addition to rational number, non-rational numbers or irrational.

$\mathbb{C}$ is the set of complex numbers, i.e. the set of real numbers $\mathbb{R}$ and all imaginary numbers $\mathbb{I}$.

87. Match the following in List - I and List - II, for a function f :

List - I

List - II

- a. $\forall x \forall y (f(x) = f(y) \rightarrow x = y)$ i. Constant
b. $\forall y \exists x (f(x) = y)$ ii. Injective
c. $\forall x f(x) = k$ iii. Surjective

Code :

- | | | | |
|-----|------------|------------|------------|
| | (a) | (b) | (c) |
| (a) | (i) | (ii) | (iii) |
| (b) | (iii) | (ii) | (i) |
| (c) | (ii) | (i) | (iii) |
| (d) | (ii) | (iii) | (i) |

Ans. (d) :

88. Which of the relations on $\{0, 1, 2, 3\}$ is an equivalence relation ?

- (a) $\{(0, 0) (0, 2) (2, 0) (2, 2) (2, 3) (3, 2) (3, 3)\}$
(b) $\{(0, 0) (1, 1) (2, 2) (3, 3)\}$
(c) $\{(0, 0) (0, 1) (0, 2) (1, 0) (1, 1) (1, 2) (2, 0)\}$
(d) $\{(0, 0) (0, 2) (2, 3) (1, 1) (2, 2)\}$

Ans. (b) : For a relation to be equivalent, it must be reflexive (xRx), Symmetric (if xRy then yRx) and Transitive (If xRy and yRz then xRz).

Option (b) is equivalence as it is reflexive, symmetric and transitive.

89. Which of the following is an equivalence relation on the set of all functions from $\mathbb{Z}$ to $\mathbb{Z}$?

- (a) $\{(f, g) \mid f(x) - g(x) = 1 \forall x \in \mathbb{Z}\}$
(b) $\{(f, g) \mid f(0) = g(0) \text{ or } f(1) = g(1)\}$
(c) $\{(f, g) \mid f(0) = g(1) \text{ and } f(1) = g(0)\}$
(d) $\{(f, g) \mid f(x) - g(x) = k \text{ for some } k \in \mathbb{Z}\}$

Ans. (b): Let A be set of all function from Z to Z.

Option A– $R_1 = \{(f, g) \mid f(x) = g(x) = 1, \forall x \dots \text{it} \in Z\}$ is not reflexive because A is set of all the functions from Z to Z. And the given condition.

$f(x) = g(x) = 1, \forall x \in Z$ will only be satisfied by an unique pair (f, f).

So, R_1 is symmetric, Transitive but not reflexive.

R_1 = Not reflexive, symmetric, transitive, not irreflexive, Antisymmet, Not Asymmetric, Not an equivalence. Relation, Not a partial order relation.

Option B– $R_2 = \{(f, g) \mid f(0) = g(0) \text{ or } f(1) = g(1)\}$

It is reflexive, symmetric but not transitive.

for e.g.– Let (f, g), (g, h) $\in R_2$ such that

$$f(0) = g(0) = 1 \text{ and } g(1) = h(1) = 2 \text{ and } h(0) = 4, f(1) = 5$$

So, we can see that (f, h) $\notin R_2$

Hence, not transitive.

R_2 = Reflexive, symmetric, not transitive, not irreflexive, not antisymmetric, not asymmetric, not an equivalence relation, not a partial order relation.

Option D– $R_4 = \{(f, g) \mid f(x) - g(x) = R, \text{ for some } K \in Z\}$

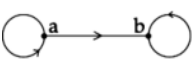
R_4 is reflexive, symmetric, transitive.

Hence, an equivalence relation.

Note– that R_4 is reflexive, symmetric, transitive because range of all the function is Z and we know that $Z - Z = Z$

R_4 = Reflexive, symmetric, transitive, not irreflexive, not antisymmetric, not asymmetric, an equivalence relation not a partial order relation.

90. Which of the following statements is true ?

- (a) $(Z, \leq)$ is not totally ordered
- (b) The set inclusion relation $\subseteq$ is a partial ordering on the power set of a set S
- (c) $(Z, \neq)$ is a poset
- (d) The directed graph  is not a partial order

Ans. (b) :

(a) $(Z, \leq)$ is not totally ordered (false)

(b) The set inclusion relation $\subseteq$ is a partial ordering on the power set of a set S (True)

$$\text{Let } S = \{a, b\}$$

$$P(S) = \{\{a\}, \{b\}, \{a, b\}, \emptyset\}$$

(c) $(Z, \neq)$ is a poset [False] because Reflexive Relation doesn't satisfy.

(d) The directed graph is not a partial order [False].

91. CMOS is a Computer Chip on the motherboard, which is :

- (a) RAM
- (b) ROM
- (c) EPROM
- (d) Auxillary storage

Ans. (a) : CMOS stands for "Complementary Metal-Oxide-Semiconductor". It's the name of a manufacturing process used to create, processors, RAM, and digital logic circuits, and is also the name for chips created using that process.

92. In RS flip-flop, the output of the flip-flop at time $(t + 1)$ is same as the output at time t , after the occurrence of a clock pulse if :

- (a) $S = R = 1$ (b) $S = 0, R = 1$
 (c) $S = 1, R = 0$ (d) $S = R = 0$

Ans. (d) : $S = R = 0$ is the normal resting condition of the flip-flop. It has no effect on the output state of the flip-flop. Both Q and $\bar{Q}$ output remain in the logic state they were in prior to this input condition.

$S = R = 0$		
R	S	Q_{t+1}
0	0	No change
0	1	0
1	0	1
1	1	* race

93. Match the terms in List - I with the options given in List - II :

List - I

- a. Decoder
 b. Multiplexer
 c. De multiplexer

List - II

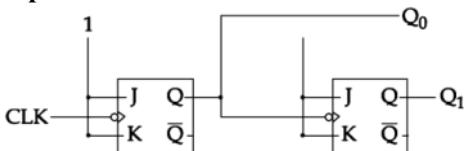
- i. 1 line to 2^n lines
 ii. n lines to 2^n lines
 iii. 2^n lines to 1 line
 iv. 2^n lines to 2^{n-1} lines

Code :

- | | | | |
|-----|------------|------------|------------|
| | (a) | (b) | (c) |
| (a) | (ii) | (i) | (iii) |
| (b) | (ii) | (iii) | (i) |
| (c) | (ii) | (i) | (iv) |
| (d) | (iv) | (ii) | (i) |

Ans. (b) :

94. What does the following logic diagram represent ?



- (a) Synchronous Counter
 (b) Ripple Counter
 (c) Combinational Circuit
 (d) Mod 2 Counter

Ans. (b) : Ripple Counter— A ripple counter is an asynchronous counter where only the first flip-flop is clocked by an external clock. All subsequent flip-flops are clocked by the output of the preceding flip-flop. Asynchronous counters are also called ripple counters because of the way the clock pulse ripples it way through the flip-flops.

95. The hexadecimal equivalent of the binary integer number 110101101 is :

- (a) D24 (b) 1 B D
(c) 1 A E (d) 1 A D

Ans. (d) : Group the given number in subgroup of 4 from right and assign alphabet when it exceeds 1001:

1010 – A 1110 – E
1011 – B 1111 – F
1100 – C
1101 – D

Given number is 110101101

1101 – D
1010 – A
1 – 1

Hexadecimal equivalent is – 1AD.

96. Perform the following operation for the binary equivalent of the decimal numbers $(-14)_{10} + (-15)_{10}$

The solution in 8 bit representation is :

- (a) 11100011 (b) 00011101
(c) 10011101 (d) 11110011

Ans. (a) : $(-14)_{10} + (-15)_{10}$

Now 29 in binary (8 bit) is 00011101. But the number is negative. Take its complement 11100010 and 1 to it i.e. – 11100011.

97. Match the items in List - I and List - II :

List - I

List - II

- | | |
|--|-----------------|
| a. Interrupts which can be delayed when a much highest priority interrupt has occurred | i. Normal |
| b. Unplanned interrupts which occur while executing a program | ii. Synchronous |
| c. Source of interrupt is in phase with the system clock | iii. Maskable |
| | iv. Exception |

Code :

- | | (a) | (b) | (c) |
|-----|-------|------|-------|
| (a) | (ii) | (i) | (iv) |
| (b) | (ii) | (iv) | (iii) |
| (c) | (iii) | (i) | (ii) |
| (d) | (iii) | (iv) | (ii) |

Ans. (d) :

98. Which of the following mapping is not used for mapping process in cache memory ?
- (a) Associative mapping
 - (b) Direct mapping
 - (c) Set-Associative mapping
 - (d) Segmented - page mapping

Ans. (d) : The three different types of mapping used for the purpose of cache memory are :

- Direct mapping
- Associative mapping
- Set-Associative mapping, segment–page mapping.
- Is not used.

99. Simplify the following using K-map :
- $F(A, B, C, D) = \sum (0, 1, 2, 8, 9, 12, 13)$
- $d(A, B, C, D) = \sum (10, 11, 14, 15)$
- d stands for don't care condition.**
- (a) $A + \overline{B}\overline{D} + BC$
 - (b) $A + \overline{B}\overline{D} + \overline{B}\overline{C}$
 - (c) $\overline{A} + \overline{B}\overline{D}$
 - (d) $\overline{A} + \overline{B}\overline{C} + \overline{B}\overline{D}$

Ans. (b) :

$\begin{matrix} & AB \\ CD \backslash \end{matrix}$	00	01	11	10
00	1		1	1
01	1		1	1
11			×	×
10	1		×	×

$A + \overline{B}\overline{D} + \overline{B}\overline{C}$

100. In 8085 microprocessor, what is the output of following program ?
- ```
LDA 8000H
MVI B, 30H
ADD B
STA 8001H
```
- (a) Read a number from input port and store it in memory
  - (b) Read a number from input device with address 8000H and store it in memory at location 8001H
  - (c) Read a number from memory at location 8000H and store it in memory location 8001H
  - (d) Load A with data from input device with address 8000H and display it on the output device with address 8001H

**Ans. (c) :** LDA 8000H (Memory location)

MVI B, 30H

ADD B

STA 8001H (Stored in memory location)

Read a number from memory at location 8000H and store it in memory location 8001H.